

3. Project Design Phase

FITBUDDY – AI FITNESS PLAN GENERATOR USING GEMINI MODELS

- **System Architecture:** The user interacts with the web interface. FastAPI receives requests and coordinates application logic. The AI service sends suitable prompts to the Gemini model. SQLAlchemy manages communication with the SQLite database. Jinja2 renders web pages.
- **Main Modules:** User Input Module; AI Plan Generator; Feedback and Plan Update Module; Nutrition and Recovery Module; Database Module; Administrator Module; Web Interface.
- **Basic Data Flow:** User Input → FastAPI → Prompt Construction → Gemini Model → Generated Plan → Database Storage → Web Presentation.
- **Database Design:** The database can maintain user records and generated workout-plan information. SQLAlchemy provides the ORM layer for database operations.
- **Interface Design:** Pages should provide clear forms for user information, generated plan display, feedback submission, and administrator viewing.